

Assignment 7

Module -7: CCNA - Routing and switching

1. Explain Switch

Ans :

- A **network switch** is a device used in a LAN (Local Area Network) to connect multiple devices like computers, printers, and servers.
- It works at Layer 2 (Data Link Layer) of the OSI model.

○ **How it works:**

- Every device has a **MAC address**
- Switch stores MAC addresses in a table (MAC table)
- When data comes → it checks destination MAC → sends only to that port

○ **Key functions:**

- ✓ Uses **MAC addresses** to forward data
- ✓ Reduces network traffic (unlike hub)
- ✓ Sends data only to the intended device (not broadcast to all)

2. Explain Switch Boot Sequence

Ans :

- ✓ When a switch starts, it follows these steps:

1. **POST (Power-On Self Test)**

Checks hardware components like CPU, RAM.

2. **Load Boot Loader**

Stored in ROM, initializes the system.

3. **Load IOS (Cisco IOS)**

Loads the operating system from Flash memory.

4. **Load Configuration File**

Loads startup-config from NVRAM.

5. **Ready State**

Switch becomes operational

3. Explain Three Methods to access Switch Command Line Interface

Ans :

1. Console Access

- Direct connection using console cable
- Used for initial configuration

2. Telnet

- Remote access over network
- Not secure (plain text)

3. SSH (Secure Shell)

- Secure remote access
- Encrypted communication (recommended)

4. Explain and Configuring the Cisco Internet Operating System

Ans :

```
enable
conf t
hostname deep
int fa 0/1
ip address 10.0.0.1 255.0.0.0
no shut
ex
do wr
line console 0
password hello
login
ex
do wr
```

- **Functions:**

- ✓ Device configuration
- ✓ Routing & switching
- ✓ Security management

5. Explain Switch Port

Ans :

- ✓ A **switch port** is a physical interface used to connect devices.

○ **Types:**

1. **Access Port** → Connects to end devices (PCs)
2. **Trunk Port** → Connects switches (carries multiple VLANs)

6. R1, R2, R3, and R4 have their Fast Ethernet 0/0 interfaces attached to the same VLAN. A network engineer has typed a configuration for each router by using a word processor. He will later copy and paste the configuration into the routers. Examine the following exhibit, which lists configuration for the four routers, as typed by the network engineer. Assuming that all four routers can ping each other's LAN IP addresses after the configuration has been applied, choose the routers that will be able to form a neighbor relationship with the other routers on the LAN. (You can assume that, if not shown in the exhibit, all other related parameters are still set to their defaults.) (Choose two)
- A. R1
 - B. R2
 - C. R3
 - D. R4
 - E. None of the routers will exchange routing information.

Ans : B r2 and C r3

7. enable secret [password] is hashed using the algorithm.
- A. MD5
 - B. AH
 - C. PSK
 - D. ESP
 - E. WPA2

Ans : A md5

8. An engineer connects to Router R1 and issues a show ip ospf neighbor command. The status of neighbor 2.2.2.2 lists FULL/BDR. What does the BDR mean?
- A. R1 is an Area Border Router.
 - B. R1 is a backup designated router.
 - C. Router 2.2.2.2 is an Area Border Router.
 - D. Router 2.2.2.2 is a backup designated router.

Ans : D Router 2.2.2.2 is a backup designated router.

9. Which command is used to view the neighbor discovery table on a PC?

- A. show ipv6 neighbor
- B. show ipv6 neighbors
- C. netsh interface ipv6 show neighbor
- D. netsh interface ipv6 show neighbors

Ans : C netsh interface ipv6 show neighbor

10. What type of variable is being shown? Routers = [R1,R2,R3]

- A. List
- B. Dictionary
- C. Simple
- D. Unsigned integers

Ans : A list

11. Identify the fields in an IPv4 header. (Choose three)

- A. Host component
- B. Time to Live
- C. Source address
- D. Destination address
- E. Network address

Ans : B time to live , **C** source address , **D** destination address